

Ryan Greene

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EDUCATION

Penn State University, State College
Bachelor of Science in Computer Engineering

Expected Graduation: May 2026
GPA: 3.04

Relevant Coursework: Electronic Circuit Design; Object-Oriented, Systems Programing, Data Structures & Algorithms, Operating Systems; Digital Hardware Design; Signal & Systems, Differentials, Several Variables, Matrices, Discrete Math; Physics: Electricity, Magnetism, Wave Motion, Quantum Mechanics; Introduction to Computer Architecture, Microprocessors, Embedded Systems, Communication Networks, Programming Models for Big Data

SKILLS

- Programming Languages:* Python, Verilog / VHDL, Java, SQL, C, C++, MIPS & x86 Assembly, JSON
- Frameworks and Tools:* PowerShell, Xilinx, Vivado, Git, Linux, NI LabView, Multisim, MATLAB, PGAdmin, AquaData Studio, AWS, DynamoDB,Gdb,Linux, PowerBI, Raspberry Pi, Postgres SQL, Jira, MapReduce

WORK EXPERIENCE

FINRA – Financial Industry Regulatory Authority — Rockville, MD
Intern – Software/Data Engineer — May 2025 – August 2025

- Built a scalable data migration pipeline and authored 10,000+ lines of production SQL (DDL, ETL, automation) in PostgreSQL, enabling seamless ingestion of millions of market regulation records.
- Owned and maintained a Power BI dashboard tracking dataset usage, adapting SQL queries and reports as datasets, columns, and schemas were retired or replaced.
- Enhanced cross-departmental data visibility by retiring legacy database structures, mapping report dependencies, and developing backend workflows for analytics, compliance, and GenAI use cases.

PROJECTS

AI-Powered Contract Change Detection with AWS Lambda & LLMs
2025 FINRA Creatathon – Team awarded 1st Place by FINRA

- Architected a serverless pipeline using AWS Lambda, DynamoDB, and Step Functions to detect and classify changes in 3rd-party vendor Terms & Conditions documents via weekly web scraping and semantic analysis.
- Implemented LLM-powered clause comparison (Claude) with JSON-based output, highlighting risk level (High/Medium/Low), and routing results to impacted teams; supported DOCX, PDF, and TXT formats.
- Engineered fault-tolerant document ingestion with hash-based deduplication, feedback loops, and real-time notifications; reduced 100+ hours of manual contract review to <25 hours through automation.

Waveform Generator with Real-Time Clock & ADC Acquisition (Assembly)

- Leveraged RTI interrupts for 24-hour clock display, and UART interface for interactive control, demonstrating proficiency in embedded systems timing, low level hardware interfacing, and real-time data acquisition.
- Engineered and debugged using CodeWarrior and HCS12 assembly to produce sawtooth/triangle/square waveforms with precise timer-driven interrupts and serial terminal commands, captured 2,048 ADC samples at 8 kHz.

Memory Management System (C)

- Implemented malloc, realloc, and free using bitwise metadata and mem_sbrk for dynamic heap management.
- Ensured 16-byte alignment for all allocations to match standard libc behavior.

Threaded Channel Synchronization System (C, pthread, Linux)

- Implemented a concurrent message-passing system supporting blocking and non-blocking communication between multiple sender and receiver threads. Used POSIX mutexes, semaphores, and condition variables to ensure synchronization and race-free data exchange. Verified correctness using Valgrind and data race sanitizers

Pipelined Processor (Verilog)

- This processor is a 32-bit, five stage (IF, ID, EXE, MEM and WB) pipelined CPU with forwarding that can handle 20 MIPS instructions. This was made using Verilog and MIPS instruction set on Xilinx and Vivado.

EXTRACURRICULARS

- Institute of Electric and Electronics Engineers (IEEE) – Penn State Chapter
 - Pillar (Thon) – Member
 - Engineering Leadership Society – Member
 - Claude Builder Club

August 2022-Present

August 2024-Present

August 2024-Present

August 2025-Present